

Section 1

1. e) The int8-quantized base model came out the smallest at 5.74 KB. I wanted to see if combining two of the techniques could push the size down further, so I applied int8 quantization on top of the KD student model. Since the student already has a smaller architecture, there are fewer weights to quantize in the first place; hence, the two methods would likely work together. The result's size was 3.62 KB with 96% accuracy, which is a 37% reduction over the previous best technique. There is, however, a small accuracy drop from 100% to 96%; however, this makes sense given that the student model was already trading some performance for a lighter model.

Section 2

1. The free tier Learning Blocks are:

Supervised

- Classification: gesture or audio recognition
- Regression: predicting a sensor reading, e.g. temperature or speed

Unsupervised

- Anomaly Detection (GMM): anomaly detector for vibration data
- Anomaly Detection (K-means): detecting unusual patterns in accelerometer data from a machine
- Anomaly Detection: custom outlier detection model to detect anomalies in environmental sensor data

2. The free tier available layers are:

- Dense: a fully connected layer where every neuron connects to every input; mainly used for learning patterns from processed data
- 1D Convolution / pooling: learns features along a single dimension; mainly used for raw time-series or audio data
- 2D Convolution / pooling: learns features across two dimensions; mainly used for image-like data or 2D DSP outputs (e.g. spectrograms)
- Reshape: converts 1D data into a multi-dimensional format, used as a "bridge" layer before feeding data into a convolutional layer
- Flatten: collapses multi-dimensional data back into a single vector; this is needed after a convolutional layer before passing to a Dense output layer (opposite of reshape layer)
- Dropout: randomly drops a fraction of connections during training to prevent overfitting; useful for when training accuracy is much higher than validation accuracy

3. The model compression options are:

- Quantized (int8): Converts model weights from float32 to 8-bit integers, reducing memory footprint and speeding up inference on edge hardware

- Unoptimized (float32): No compression, keeps the original full-precision weights; useful as a baseline to compare accuracy against the above version
4. The input modalities supported are:
- Image
 - Black Forest Labs Flux-Pro Synthetic Image Generator (Replicate.com's API)
 - DALL-E 3 Synthetic Image Generator (OpenAI)
 - Audio/Voice
 - Whisper Synthetic Voice Generator (OpenAI's text-to-speech model)
 - 3D/Simulation
 - NVIDIA Omniverse

Synthetic data is important in machine learning applications, as real-world data collection is often expensive, time-consuming, or impractical. This is especially the case for edge devices where you might need thousands of labeled examples of rare events like equipment failures or specific gestures. Synthetic data can fill in gaps in the training set, improve class balance, and let you generate edge cases that would be hard to capture naturally. Additionally, it can do so without the privacy concerns associated with real user data.